

Climate Science: Mechanisms, Evidence, and Projections

1. The Greenhouse Effect

The greenhouse effect is the warming that results when the atmosphere traps heat radiating from Earth toward space. Certain gases in the atmosphere (greenhouse gases) block heat from escaping. Life on Earth depends on the greenhouse effect — without it, Earth's average temperature would be about -18°C instead of the current 14°C .

The primary greenhouse gases are: water vapor (H_2O), which is the most abundant and acts as a feedback amplifier; carbon dioxide (CO_2), which is the most important long-lived greenhouse gas and has increased from 280 ppm pre-industrial to over 420 ppm in 2023; methane (CH_4), which is 80 times more potent than CO_2 over 20 years; nitrous oxide (N_2O), mainly from agriculture; and fluorinated gases, synthetic and extremely long-lived.

The enhanced greenhouse effect refers to the additional warming caused by human activities that increase the concentration of greenhouse gases. Since industrialization began around 1850, human activities — primarily fossil fuel combustion, deforestation, and agriculture — have increased atmospheric CO_2 by over 50%.

2. Evidence for Anthropogenic Climate Change

Multiple independent lines of evidence confirm that Earth's climate is warming and that human activities are the primary driver since the mid-20th century:

Temperature Records: NASA and NOAA maintain global surface temperature records showing that Earth's average surface temperature has increased by approximately 1.1°C since the pre-industrial period (1850-1900). The 10 warmest years on record have all occurred since 2010. The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6, 2021) states that it is 'unequivocal' that humans have warmed the climate.

Isotopic Fingerprinting: Fossil fuels have a distinct carbon isotope signature (depleted in C-13 and C-14). The decreasing ratio of C-13 to C-12 in atmospheric CO_2 over time proves that the increase is from fossil fuel burning, not natural sources.

Ocean Heat Content: The oceans have absorbed over 90% of the excess heat trapped by greenhouse gases. Ocean heat content has increased every decade since the 1960s.

Sea Level Rise: Global mean sea level has risen approximately 20 cm since 1900, with the rate accelerating since the 1990s. Both thermal expansion of warming ocean water and melting of

glaciers and ice sheets contribute.

Arctic Sea Ice: Arctic summer sea ice extent has declined by about 40% since satellite observations began in 1979. The Arctic is warming approximately 4 times faster than the global average (Arctic amplification).

Glacier Retreat: Almost all of the world's glaciers are retreating. The World Glacier Monitoring Service tracks over 5,000 glaciers globally, with mass balance data consistently negative since the 1980s.

3. Climate Feedback Mechanisms

Climate feedbacks are processes that amplify or dampen the initial warming caused by greenhouse gas increases:

Ice-Albedo Feedback (Positive): Ice and snow are highly reflective (high albedo). As temperatures rise, ice and snow melt, exposing darker ocean or land surfaces that absorb more sunlight, causing further warming. This is a major factor in Arctic amplification.

Water Vapor Feedback (Positive): A warmer atmosphere holds more water vapor, which is itself a greenhouse gas. This approximately doubles the warming from CO₂ alone and is the largest positive feedback in the climate system.

Lapse Rate Feedback: As the atmosphere warms, the temperature gradient between the surface and upper atmosphere changes, affecting how much heat is radiated to space.

Cloud Feedbacks (Mixed): Clouds have two opposing effects — they reflect incoming sunlight (cooling) and trap outgoing longwave radiation (warming). How cloud cover and properties change with warming is the largest source of uncertainty in climate projections. Current evidence suggests a net positive feedback from clouds.

Permafrost Carbon Feedback (Positive): About 1.5 trillion tonnes of carbon are stored in frozen Arctic soils. As permafrost thaws, microbes decompose this organic matter, releasing CO₂ and methane — potentially triggering a significant feedback loop.

Carbon Cycle Feedbacks: As oceans warm, they absorb less CO₂ from the atmosphere (reduced solubility). Forest dieback from heat, drought, and pests can convert carbon sinks into carbon sources.

4. IPCC Projections and Emissions Scenarios

The IPCC uses Shared Socioeconomic Pathways (SSPs) to model different future scenarios based on greenhouse gas emissions trajectories and societal choices:

SSP1-1.9 (Very Low Emissions): Aggressive mitigation with net-zero CO₂ by ~2050. Global warming likely limited to 1.0-1.8°C by 2100. Requires rapid transformation of energy, land, and industrial systems.

SSP2-4.5 (Intermediate): Current policies trajectory with some mitigation. Warming of 2.1-3.5°C by 2100. Sea level rise of 0.3-0.6m by 2100.

SSP5-8.5 (Very High Emissions): Business as usual with high fossil fuel use. Warming of 3.3-5.7°C by 2100. Sea level rise of 0.6-1.0m by 2100 (higher if ice sheet instabilities triggered). This scenario was previously considered the 'worst case' and is now regarded as the upper bound of plausible outcomes given current trends.

The Paris Agreement (2015) aims to limit warming to well below 2°C above pre-industrial levels, preferably 1.5°C. Current nationally determined contributions (NDCs) are insufficient to meet the 1.5°C target; they imply warming of approximately 2.5-3°C.

5. Impacts of Climate Change

Climate change is already causing wide-ranging impacts across natural and human systems:

Extreme Weather: Climate change increases the frequency and intensity of heatwaves. Heavy precipitation events have increased in most regions. Atlantic hurricanes show increasing intensity and rapid intensification. Droughts are more severe in many areas.

Biodiversity and Ecosystems: Coral bleaching events (caused by ocean warming and acidification) have devastated reefs worldwide — the Great Barrier Reef experienced mass bleaching in 2016, 2017, 2020, 2022, and 2024. Roughly 1 million species face extinction risk partly from climate change. Species are shifting their ranges toward the poles and to higher elevations.

Food and Water Security: Crop yields of wheat, rice, and maize are projected to decline by up to 25% with 2°C warming in tropical regions. Glacier retreat threatens freshwater supplies for hundreds of millions of people in the Hindu Kush-Himalayan region. Ocean acidification (pH has dropped from 8.2 to 8.1 since industrialization, a 26% increase in acidity) threatens shellfish and marine food webs.

Human Health: Heat-related mortality is rising, particularly for elderly populations. Expanded range of vector-borne diseases (malaria, dengue). Air quality worsens with increased wildfire smoke and ozone formation. Mental health impacts from climate disasters and displacement are increasingly documented.

Sea Level Rise and Displacement: At 1°C warming, low-lying coastal cities like Miami, Mumbai, and Shanghai face increasing flood risk. Up to 1 billion people could be displaced by 2050 due to sea level rise, extreme heat, and drought if emissions are not drastically reduced.

6. Mitigation Strategies

Mitigation refers to actions to reduce greenhouse gas emissions or enhance carbon sinks:

Energy Transition: Replacing fossil fuels with renewable energy is central to mitigation. Solar photovoltaic costs have fallen by 90% since 2010, and wind costs by 70%. Renewables now account for over 30% of global electricity generation. Full decarbonization of the power sector is technically achievable with current technologies by 2050.

Energy Efficiency: More efficient buildings, vehicles, and industrial processes can reduce emissions by 40-70% across sectors. Heat pumps (for space heating and cooling) are 2-4 times more efficient than combustion heating.

Electrification: Replacing fossil fuel end uses (cars, heating, industry) with electricity allows decarbonization once the electricity grid is clean. Electric vehicle (EV) sales hit 10 million globally in 2022.

Carbon Capture and Storage (CCS): Capturing CO₂ from power plants or industrial processes and storing it underground. Currently expensive and limited in scale, but essential for hard-to-abate sectors.

Nature-Based Solutions: Protecting and restoring forests, wetlands, and soils can sequester significant carbon. Forests absorb approximately 2.6 billion tonnes of CO₂ per year globally. However, nature-based solutions alone cannot compensate for insufficient emissions reductions.

Carbon Pricing: Putting a price on carbon emissions through carbon taxes or emissions trading schemes (ETS) is considered one of the most economically efficient mitigation tools. Over 70 carbon pricing mechanisms now exist globally, covering about 23% of global emissions.

7. Tipping Points and Irreversibility

Climate tipping points are thresholds in the climate system that, once crossed, lead to self-reinforcing changes that may be irreversible on human timescales:

West Antarctic Ice Sheet (WAIS) Collapse: The WAIS is a marine ice sheet resting on bedrock below sea level. Warm ocean water is melting it from below (marine ice sheet instability). If destabilized, it could contribute 3-5 meters of sea level rise over centuries to millennia. Some scientists believe it may already be past a tipping point.

Greenland Ice Sheet: Holds enough ice to raise sea levels by 7 meters. Research suggests its tipping point may be between 1.5-2°C of global warming, meaning its eventual loss is already committed even if warming stabilizes at 1.5°C.

Amazon Dieback: Deforestation (currently 20% of the Amazon has been cleared) combined with climate change could push the Amazon past a tipping point around 20-25% deforestation, causing dieback of large portions of the rainforest and converting it from a carbon sink to a carbon source.

Atlantic Meridional Overturning Circulation (AMOC): This ocean circulation system transports warm water northward, moderating European climate. Evidence suggests it is already weakening; its collapse would cause dramatic cooling in Europe and disruption to global weather patterns.

Boreal Forest Dieback: The boreal forests of Canada and Russia are experiencing warming at twice the global rate, increasing fire frequency and bark beetle outbreaks.

8. Summary

Climate change is the defining challenge of the 21st century. The scientific evidence is unequivocal: Earth is warming due to human greenhouse gas emissions, impacts are already severe and widespread, and further warming will intensify these impacts. Limiting warming to 1.5°C is technically possible but requires an immediate and transformative reduction in global emissions across all sectors. The window for action is narrowing rapidly — every fraction of a degree of avoided warming significantly reduces the risk of crossing irreversible tipping points.